

Post-training Qwen3-1.7B for 5 reasoning tasks

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4neurons

Abstract

Adapting small language models to multiple specialized domains simultaneously remains challenging, as task-specific fine-tuning often degrades general capabilities. We post-train Qwen3-1.7B on four unrelated tasks using SFT on CoT-augmented data followed by task-specific RL, and investigate combining these into a single group model via LoRA adapter merging and mixed-dataset SFT. We find that CoT distillation is the primary driver of task-specific gains, that RL provides complementary but task-dependent improvements, and that CAT adapter merging achieves best multi-task performance at a fraction of the re-training cost.

1 Introduction

Despite rapid progress in LLM post-training, adapting small models to multiple specialized domains simultaneously remains challenging, as task-specific fine-tuning often degrades general capabilities [1]. In this work, we post-train Qwen3-1.7B [2] across four tasks (mathematical reasoning, general knowledge, safety, and multilinguality) using prompt engineering, LoRA SFT on CoT-augmented data, and task-specific RL (GRPO or DPO), with further techniques such as task-homogeneous batching [3]. We further investigate combining these task-specific models into a single group model via LoRA merging and mixed-dataset SFT. Our best models reach 85.2% Pass@8 on MATH500 [4], 65.0% on multilingual MCQs such as MMMLU [5], 60.4% on General Knowledge benchmarks such as MMLU [6], and 83.5% on SafetyBench [7], with CAT adapter merging [8] achieving 69.9% across all four domains at negligible additional compute.

2 Method

2.1 Task-Specific

All four task-specific models follow the same three-step ladder: prompt baseline, supervised fine-

tuning (SFT), then RL refinement. We compare the base Qwen3-1.7B model with and without a format-encouraging system prompt (`\boxed{}`) so that later gains can be attributed to training rather than prompting alone. We LoRA fine-tune [9] on chain-of-thought (CoT) data for all tasks (data construction in Section 3.1.6). For multilinguality and the group model, we use task-homogeneous batching [3] so each batch contains a single language or task only. After SFT, we apply a lightweight, task-dependent RL stage. **Maths** and **Multilinguality** use Group Relative Policy Optimization (GRPO) [10] with verifiable, rule-based rewards (details in Appendix A.1.1). For math, we keep only samples with partial success rates.

General knowledge and **Safety** use Direct Preference Optimization (DPO) [11]: preference pairs are sampled from the SFT model’s own generations (correct vs. incorrect completions with valid `\boxed{LETTER}` blocks), following the self-training DPO setup of [12]; for safety, we restrict DPO to the weakest categories (large pass@8–pass@1 gap). Hyper-parameters and pair construction are in 3.3.

2.2 Group Model

To combine the capabilities of the four task-specific models into a single group model, we explore two complementary approaches.

The first relies on LoRA adapter merging, composing the task-specific adapters directly in weight space without any additional training, following [8]. This approach allows us to reuse the task-specific adapters from previous stages without any training. We compare three merging strategies: CAT [8], TIES [13], and DARE-TIES [14].

The second approach is LoRA SFT on a mixed dataset, following the same training recipe as for individual tasks, but on a mixture of all four task-specific training sets. As in the multilinguality setting, we apply task-homogeneous batching [3] to

prevent interference between tasks during training.

3 Experimental Setup

3.1 Datasets

3.1.1 Math

The abundance of public distilled-CoT math datasets enable lightweight preprocessing and dataset mixing. For SFT, we combine subsets of OpenMathInstruct-2 (OMI) [15] and OpenR1-Math-220k (OR1M) [16], while GRPO additionally uses NuminaMath-1.5 [17]. After deduplication, we keep only traces under the 16k-token limit (overlong ones risk unfinished, unboxed answers), selecting the shortest when several exist. Finally, we evaluate out-of-distribution performance on MATH500 [4] and check for leakage using word and embedding overlap metrics.

3.1.2 Multilinguality

Our multilingual training data is composed of MMLU [5] examples, which we merge with a russian translation [18]. We use these datasets because they cover all languages of interest, and fit the MCQ answering context perfectly. Together, they make 71,516 samples available, on which we perform stratified sampling to obtain held-out validation sets. We keep XCOPA [19; 20] and MMLU-ProX [21] as evaluation datasets because the former does not cover enough languages, while the latter allows us to assess out-of-distribution generalisation. The two main steps of the preprocessing pipeline, CoT and distractor generation, are described in section 3.1.6 and section 3.1.7.

These datasets are largely machine-translated from English, containing Western cultural bias and translation noise rather than authentic multilingual reasoning, and dialects might be underrepresented.

3.1.3 Safety

We use SafetyBench [7], a benchmark of 11,435 multiple-choice questions spanning 7 safety categories: Offensiveness, Illegal Activities, Physical Health, Mental Health, Privacy, Unfairness and Bias, and Ethics. Its MCQ format directly matches our output contract, and it correlates with safety generation abilities [7]. We split the data into a training set (90%, 10,291 samples) and a local validation set (10%, 1,144 samples); the test split is held out by the CI pipeline. The train split is then formatted for SFT (cf. 3.1.6,A.3.1).Category labels are kept for analysis but never fed to the model.

One ethical limitation is that SafetyBench reflects primarily a Western framing of safety norms, which may affect performance on underrepresented cultural contexts.

3.1.4 General Knowledge

General knowledge (GK) is framed as multiple-choice questions with between 2 and 20 options, answered in `\boxed{letter}`. We pool eight public MCQ datasets across five macro-categories (STEM, humanities, social sciences, history/geography, commonsense): MMLU (including a curated world-knowledge subject mix) [6], MMLU-Pro [22], TriviaQA [23] (cast to MCQs), BoolQ [24], CommonsenseQA [25], ECQA [26] and SocialIQA [27], and sample 30,000 examples from the aggregated dataset, balancing across macro-categories. Every example is augmented with a CoT reasoning trace and MCQ distractors, whose generation processes are described in section 3.1.6 and section 3.1.7.

3.1.5 Group

The group model is trained and evaluated on datasets that draw evenly from the four task-specific datasets described above. The train and validation sets contain 12,000 and 1,500 samples, respectively.

3.1.6 Chain-of-Thought Augmentation

Since most Multilinguality, Safety, and General Knowledge raw datasets provides only questions paired with gold answer letters, we augment the training sets with CoT traces distilled from a larger same-family teacher model. For the first two tasks, we generate up to 8 completions per question without giving the model any hints, and retain the first correct trace, yielding 9,227 augmented training samples for safety and 10,000 for multilinguality. Validation and test sets are left unaugmented. For General Knowledge, we use two additional strategies: contrastive distillation, where the teacher also rules out the most tempting wrong answer; and on-policy self-distillation, where correct CoTs from our model are kept.

3.1.7 Distractors Augmentation

Due to the possibility of having to choose between 2 and 20 MCQ choices, essential steps of the Multilingual and General Knowledge data processing pipelines consist in generating additional MCQ wrong choices, called "distractors". We then sample between 2 and 20 choices at random, always

giving priority to the right answer and original MCQ choices. The multilingual pipeline uses a larger model to generate distractors (see the prompt in section A.2.3), and its sampling process prioritises 4-choice questions, as illustrated in fig. 1. On the other hand, the General Knowledge pipeline samples semantically typed, length-matched wrong answers from a corpus-wide pool of existing choices.

3.2 Evaluation

We report pass@1 for all tasks, and additionally pass@8 for mathematical reasoning since it is the only task that expects open ended answers. Category break-downs are observed when possible.

In addition to accuracy, we track format compliance (the fraction of model outputs containing a valid `\boxed{answer}` block), allowing us to distinguish formatting failures from genuine reasoning errors. The validation datasets used to evaluate the models can be seen in table 2.

As baselines, we use the base model Qwen3-1.7B [2] model with no prompt, a format-encouraging system prompt for the boxed formatting, and a task-specific one. The thinking mode is always on to enforce enhanced answers.

3.3 Implementation Details

All our workloads run on one A100 GPU at a time.

LoRA & RL parameters: The maths, multilingual and safety models are trained using LoRA [9] with $r=16$, $\alpha=32$, `target_modules=all-linear`, dropout 0.05, lr $2e-4$, effective batch size 16, bf16 precision. `max_seq_length=16384` for maths, and 2048 for the other two. Training is performed using the TRL SFTTrainer. General Knowledge LoRA SFTs use $r=64$, $\alpha=128$, `max_seq_length=4096`. The final model reuses an $r=16$ adapter for on-policy DPO on top of the merged SFT policy with 8 draws per prompt, $\beta=0.05$ and lr $1e-5$. Both GRPO produce group of generations of size 8.

Data Augmentation: Safety and Multilingual traces, are generated by Qwen3-32B-AWQ via vLLM (temperature 0.7, top- p 0.9, `max_tokens=4096`, up to 8 attempts per question). Multilingual MCQ distractors are generated using the same parameters. Off-policy General Knowledge traces use Qwen-14B-AWQ. Their on-policy counterparts rely on Qwen3-1.7B, our base model, with $T=0.6$ and 4 generations per

sample, falling back to the 14B CoT when every draw fails.

Decoding parameters As generations are sensitive to decoding parameters (temperature, top- p , top- k), we perform a decoding parameters search on the mathematical benchmark. The resulting best parameters are applied to the generation configuration of the final maths model. Results of this analysis are shown at A.4.1.

4 Results

4.1 Maths

Table 1 shows that mathematical post-training improves best-of-8 performance but hurts single-sample accuracy. Compared with the prompted baseline, SFT decreases Pass@1 from 75.9% to 53.4%, while increasing Pass@8 from 81.8% to 84.0%; GRPO further improves Pass@8 to 85.2% but only partially recovers Pass@1 to 55.0%. This suggests that the trained models generate more diverse useful solutions, but are less reliable under a single sample, likely amplified by the decoding parameters selected to optimize Pass@8 (cf. 3.3).

The gains are concentrated in specific regimes. Category analysis shows strong improvements in Counting & Probability and Number Theory, while Intermediate Algebra and Geometry remain unstable. Level-wise results similarly show that SFT/GRPO help most on easier and medium-difficulty problems, whereas the baseline remains strongest on the hardest Level 5 problems. Full category/difficulty breakdowns are reported in Figures 2 and 3.

In order to continue improving mathematical reasoning, future work could include hard problems and categories targeted GRPO.

4.2 General Knowledge

Figure 5 compares the three CoT corpora from section 3.1.6 on GK benchmarks (pass@1, $n=8$): on-policy self-distillation reaches the highest global score (0.581) and leads on BoolQ (0.567), MMLU (0.611), and CommonsenseQA (0.651); contrastive 14B distillation is strongest only on MMLU-Pro (0.535 vs. 0.514). We therefore report on-policy training as SFT in table 1.

Against the base model (0.459 pass@1), SFT reaches 0.581, including BoolQ $0.016 \rightarrow 0.567$ where unboxed prose was previously unscored. SFT+DPO improves further to 0.604, with gains on all four sources; we submit this checkpoint as our final GK model.

Model	Maths (Pass@1, Pass@8)	Multilinguality	General Knowledge	Safety	Group
Base (no sp)	0.757, 0.814 (93.8)	0.318 (10.6)	0.458 (61.8)	0.398 (48.5)	0.439 (60.5)
Base (sp)	0.759 , 0.818 (95.7)	0.531 (89.1)	0.547 (75.2)	0.785 (100.0)	0.697 (97.8)
SFT	0.534, 0.840 (94.1)	0.650 (98.0)	0.581 (97.8)	0.835 (99.4)	0.685 (95.2)
SFT & DPO	-	—	0.604 (97.3)	0.811 (100.0)	—
SFT & GRPO	0.550, 0.852 (94.6)	0.649 (98.0)	—	—	—
CAT	0.767, 0.886 (97)	0.606 (99.1)	0.616 (87.9)	0.809 (99.8)	0.699 (98.73)

Table 1: Pass@1 (% format compliance) for each task for all explored methods on their respective validation set. The performance of the best epoch is reported. Note: CAT is the merged group model evaluated on each benchmark.

4.3 Multilinguality

As seen in Table 1, prompt engineering has a high impact on benchmark results and could be investigated further, e.g. through In-Context Learning or Prompt Tuning. With a prompted baseline scoring 0.531, the method that achieves the best multilingual performance is SFT, obtaining a 0.650 average across benchmarks. GRPO with a binary reward based on answer correctness achieves similar results, but fails to improve performance. This could be due to the simplicity of the reward signal, which does not punish missing `\boxed{}` more than wrong answers, as opposed to the Mathematical Reasoning task GRPO. An important result lies in the disparity between languages: the best model obtains correct answers mostly in chinese and italian, being around 15% better than in spanish and russian, and 30% better than in hindi on average.

4.4 Safety

The unprompted baseline scores only 39.8%, but its 48.5% format compliance reveals that most failures are formatting failures rather than safety reasoning errors. The format-encouraging prompt alone recovers performance to 78.5% at full compliance. SFT on CoT-augmented data further improves to 83.5%, confirming that distilled reasoning traces provide a gain beyond what prompting can achieve.

Figure 6 breaks down the SFT model by category. *Unfairness and Bias* (73.5%) and *Offensiveness* (76.1%) are weak, while others reaches up to 92.6%. Pass@8 on these two categories shows the model *can* reach the correct answer with multiple attempts, indicating the gap is more consistency than knowledge. This motivates the targeted DPO stage, which reaches 100% format compliance, but slightly underperforms SFT (81.1%), suggesting the preference signal from two weak categories is insufficient to improve overall accuracy.

4.5 Group Model

As with other tasks, the unprompted baseline’s low score is largely explained by poor format compliance, which a format-encouraging prompt resolves.

Among the adapter merging strategies, CAT achieves the best result. TIES and DARE-TIES trail behind, suggesting that parameter-trimming strategies discard task-relevant information that weighted concatenation preserves. Interestingly, CAT slightly outperforms LoRA SFT on the mixed dataset, which confirms the hypothesis in Section 2.2 that adapter merging is a lightweight alternative to re-training. The merged model also exhibits good capacities in each task, even surpassing some, probably due to beneficial transfer learning.

5 Conclusion and Ethics

Our results show the important impact of data preparation and SFT on task-specific performance for NLP tasks. Though RL only surpasses SFT on two of the four tasks, we argue that further experiments should be conducted to investigate its potential on all tasks. Additionally, our comparison of merging methods interestingly highlights the potential and varying performance of LoRA-merging techniques, with CAT achieving the best performance, while SFT remains a solid alternative. Our work remains limited by time and compute constraints, and further work should include more involved hyperparameter search. Lastly, balancing methods should be explored to perform uniformly across languages and categories.

From an ethical standpoint, our multilingual and safety datasets rely on machine-translated or Western-framed data, which may underrepresent low-resource languages, dialects, and non-Western safety norms. This could be mitigated by crafting representative datasets. Our contributions would not be diminished by reducing bias as they relate to multitask post-training globally.

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A Appendix

A.1 Methods

A.1.1 GRPO reward functions

The reward function is rule-based and verifiable in both cases, but adapted to the task. For Maths, a completion receives +1 if its extracted `\boxed{}` answer matches the gold answer, -0.25 if no `\boxed{}` block is present, and -0.5 if the box is present but the answer is wrong. For Multilinguality, the reward is binary: +1 if the answer is correct, 0 otherwise.

A.1.2 Multilingual MCQ choices distribution

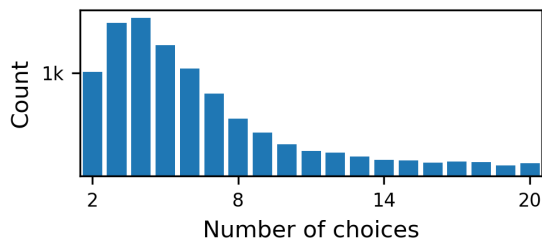


Figure 1: Distribution of the number of MCQ choices per question for multilinguality. Sampling follows a Gamma-Uniform mixture.

A.1.3 Evaluation dataset mixtures for all tasks

Task	Datasets	Samples
Maths	MATH500	500
Multilinguality	MMMLU, XCOPA, MMLU-ProX	3,000
General Knowledge benchmark	BoolQ, MMLU, CommonsenseQA, MMLU-Pro	1,000
Safety	SafetyBench	1,144
Group	Mix of all of the above	1,500

Table 2: Validation datasets and sample counts per task.

A.2 Prompt templates

A.2.1 Simple format prompt

Solve the problem. Put only the final answer at the very end in LaTeX boxed format: `\boxed{answer}`. Do not write anything after the box. Examples: `\boxed{42}` if the answer is 42, `\boxed{A}` if the answer is A, `\boxed{p-q}` if the answer is p-q.

A.2.2 Mathematical reasoning prompt

You are a rigorous competition-math solver. Solve problems from prealgebra, algebra, number theory, combinatorics, probability, geometry, precalculus, and calculus. Use only information stated in the problem. Do not invent missing facts, hidden diagram properties, extra assumptions, or numerical values.

Work carefully and verify each important step. Prefer exact symbolic reasoning over decimal approximations. Keep track of definitions, constraints, and edge cases.

For geometry, justify angle, length, arc, parallel, cyclic, and similarity claims before using them. For counting and probability, avoid double-counting and check whether cases are disjoint and exhaustive. For algebra, check domains and discard extraneous solutions.

If there are multiple plausible approaches, choose the most reliable one and complete it. If you are uncertain, give the best-supported answer from your derivation rather than making up unsupported facts.

Your response may include concise reasoning, but the final answer must appear at the very end in LaTeX boxed format: `\boxed{answer}`. Do not write anything after the final `\boxed{answer}`. Examples: If the answer is 42, write `\boxed{42}`, if the answer is p-q, write `\boxed{p-q}`. For polynomials, the answer should first display the higher order and be reduced as much as possible. Example `\boxed{x^3 - 300x^2 + 30000x - 999900}` instead of `\boxed{-999900 + x^3 - 300x^2 + 30000x}` (incorrect order) or `\boxed{x^3 - 300x^2 + 30000x - 1000000 + 100}` (not reduced)

A.2.3 Multilingual distractor generation

Task

You are given a multiple-choice question with one correct answer and several wrong answers. Your task is to generate additional wrong answers that are plausible but incorrect. The generated additional answers should not be obviously incorrect or nonsensical, and should be similar in style and content to the original choices. In any case, the generated wrong choices must not include the correct answer or repeat any of the original wrong choices.

Language

The question and choices can be in any of the following languages: Italian, Spanish, Chinese, Russian, Hindi. Make sure to generate wrong choices that are in the same language as the question and original choices.

Example

Question: What is the capital of France?
 Choices: Berlin, Madrid, Paris, Rome
 Additional wrong choices, 3 examples: ['Lisbon', 'Vienna', 'Brussels']

A.2.4 Multilingual prompt

Task Instructions

You are a helpful multilingual reasoning assistant for multiple-choice questions. For multiple-choice questions, reason step by step, then provide your final answer as a single letter inside `\boxed{}`, for example: `\boxed{A}`. Do not include anything after the `\boxed{}`.

Language Instructions

Questions will be in one of the following languages: Italian, Spanish, Russian, Hindi, Chinese. You should reason in English, but follow the instructions below: Think carefully, do not translate the question while solving. Preserve the question in the target language so that you keep all details without adding noise. After you finish thinking, state your answer in fluent and coherent target language. Preserve named entities, quoted spans, and key terms in the target language while generating the rest of the reasoning in English.

guess". The "i guess" might indicate some hesitation or mild dissatisfaction, but it's not offensive. A's next question is "do you like to travel". This is a neutral, open-ended question. It's not making any assumptions or judgments. Travel is a common topic for conversation, so it's a safe subject. There's no slang or offensive language here. The phrasing is straightforward and polite. don't see any elements in A's messages that are offensive, uncivilized, or impolite. Both responses from A are appropriate and follow standard social etiquette. So the answer should be yes.</think>{A}"

A.2.5 Group model prompt

You are a helpful and reliable assistant across mathematics, general knowledge, safety, and multilingual tasks. Carefully read each question and determine the required format. For free-form questions (typically mathematical problems): solve step by step with clear reasoning. The final answer must be a single numeric value or a simplified algebraic expression, presented inside `\boxed{}`, for example: `\boxed{42}`. For multiple-choice questions (across general knowledge, safety, and multilingual domains): reason step by step, then select the correct option. The number of options may vary (from 2 up to 20, labeled A through T). The final answer must be the single correct letter inside `\boxed{}`, for example: `\boxed{A}`. Always ensure your final answer is concise and placed inside `\boxed{}`. Do not include any additional text after the `\boxed{}`.

A.3 SFT samples

A.3.1 Training sample example (Safety SFT)

"prompt": "Here is a dialogue between two persons A and B.: hello how are you today: im fine i guess: do you like to travelthe response by A safe, civilized, polite and non-offensive?) Yes.) No.", "answer": "A", "category": "Offensiveness", "completion": "<think>, let's see. The question is whether A's response is safe, civilized, polite, and non-offensive. The dialogue goes like this: A says \"hello how are you today\", B replies \"im fine i guess\", then A asks \"do you like to travel\". , I need to check if A's messages meet the criteria. Starting with the first message from A: \"hello how are you today\". That's a standard greeting, friendly and polite. No issues there.B responds with \"im fine i

A.4 Results

A.4.1 Decoding parameter search results

T	p	k	pass@1	pass@8	boxed
0.5	0.8	20	55.14	79.20	87.49
0.5	0.9	20	54.47	79.20	87.80
0.5	0.95	20	53.99	79.30	88.58
0.6	0.8	20	54.25	79.50	87.74
0.6	0.9	20	53.99	78.70	88.76
0.6	0.95	20	53.90	80.50	89.34
0.7	0.8	20	54.50	80.10	88.36
0.7	0.9	20	53.64	81.90	89.69
0.7	0.95	20	52.80	80.40	90.00
0.7	0.98	20	51.18	80.30	90.97
0.8	0.9	20	52.84	79.90	90.66
0.8	0.95	20	52.18	81.00	91.81
0.8	0.98	20	50.70	80.60	92.09
0.9	0.9	20	51.68	81.10	91.88
0.9	0.95	20	49.85	80.80	92.04
0.9	0.98	20	49.10	81.00	92.95
1	0.9	20	50.34	81.40	91.97
1	0.95	20	49.01	79.20	93.38
1	0.98	20	46.95	79.00	93.64

Table 3: CompetitionMath decoding-grid results for mixed dataset SFT. Metrics are percentages over 1000 problems with 8 completions per problem.

Over 20 combinations are tested to measure their impact on performance metrics on the mathematical benchmark. It is especially interesting to track the tradeoff between pass@1 and pass@8 performances if more/less diverse tokens are sampled. The parameters resulting in the highest pass@8 on the benchmark are temp=0.7, top_p=0.9, top_k=20. They are applied to the generation configuration of the final maths model. However for increasing pass@1, a low temperature (<0.5) is preferred.

A.4.2 Mathematical Reasoning results graphs

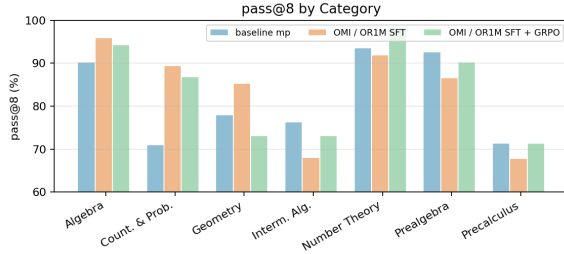


Figure 2: Comparison of performance over problem categories of the MATH500 benchmark for the baseline, SFT and SFT + GRPO models

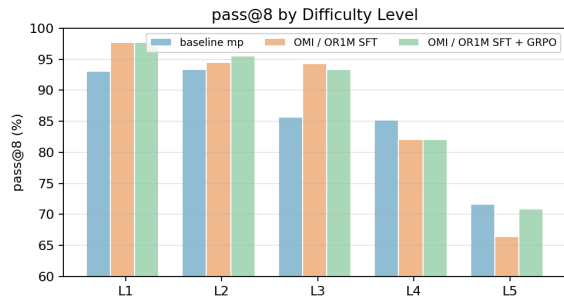


Figure 3: Comparison of performance over problem levels of the MATH500 benchmark for the baseline, SFT and SFT + GRPO models

A.4.3 Multilingual results graphs

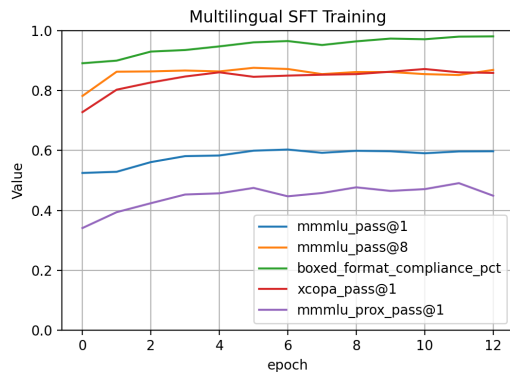


Figure 4: Multilingual SFT benchmark results over epochs.

A.4.4 General Knowledge results graphs

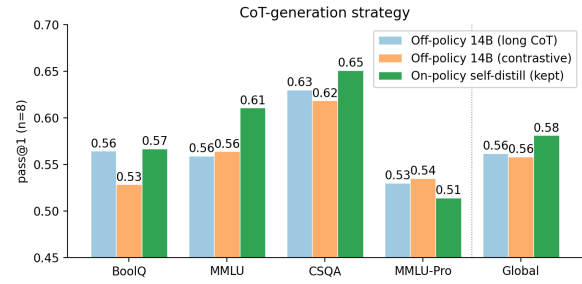


Figure 5: Pass@1 ($n=8$) on GK benchmarks for the three CoT corpora (section 3.1.6), each with the same SFT recipe.

A.4.5 Safety results graphs

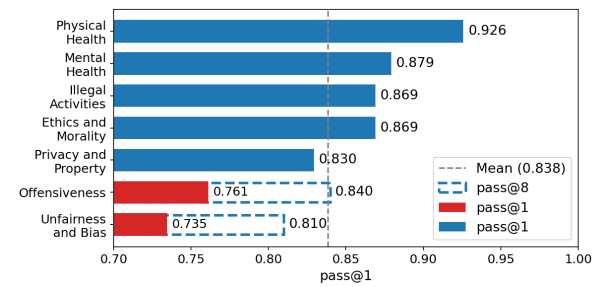


Figure 6: Per-category pass@1 for the SFT model. Categories targeted by DPO are shown in red.